Practical - 4

Practical Titleb: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

A . Theory

Add dynamic UI components such as collapsible FAQ, modal popup,bimage/content slider, notification banner, hamburger menu, and light/darkbtheme switcher using JavaScript.

1 How are DOM elements selected and modified?

DOM elements are selected using methods such as getElementById(), querySelector(), and querySelectorAll(). Their content, styles, classes, or attributes are modified using properties like textContent, innerHTML, style, and classList.

2 Are event listeners attached correctly?

Yes. Event listeners are attached using addEventListener(). Events such as click, submit, and input are handled so that the required JavaScript functions run when the user interacts with the page.

3 Is localStorage used for remembering theme choice?

Yes. localStorage can store the user's selected light/dark theme. When the page is opened again, JavaScript reads the saved value and applies the same theme automatically.

4 Does interactivity improve usability without breaking accessibility?

Yes. Features such as the hamburger menu, FAQ, modal, slider, notification banner, and theme switcher make the website easier to use. Accessibility should be maintained by using proper buttons, keyboard-accessible controls, readable text, and appropriate labels.

B. Implementation/ Output/Result

Here is my GitHub repo link the codes are here in repo:
https://github.com/dhyaani19/StudentHub-Portal.git

C.. Evidence of Work (Screenshots)

Figure 4.1 : Profile page (peofile.html)

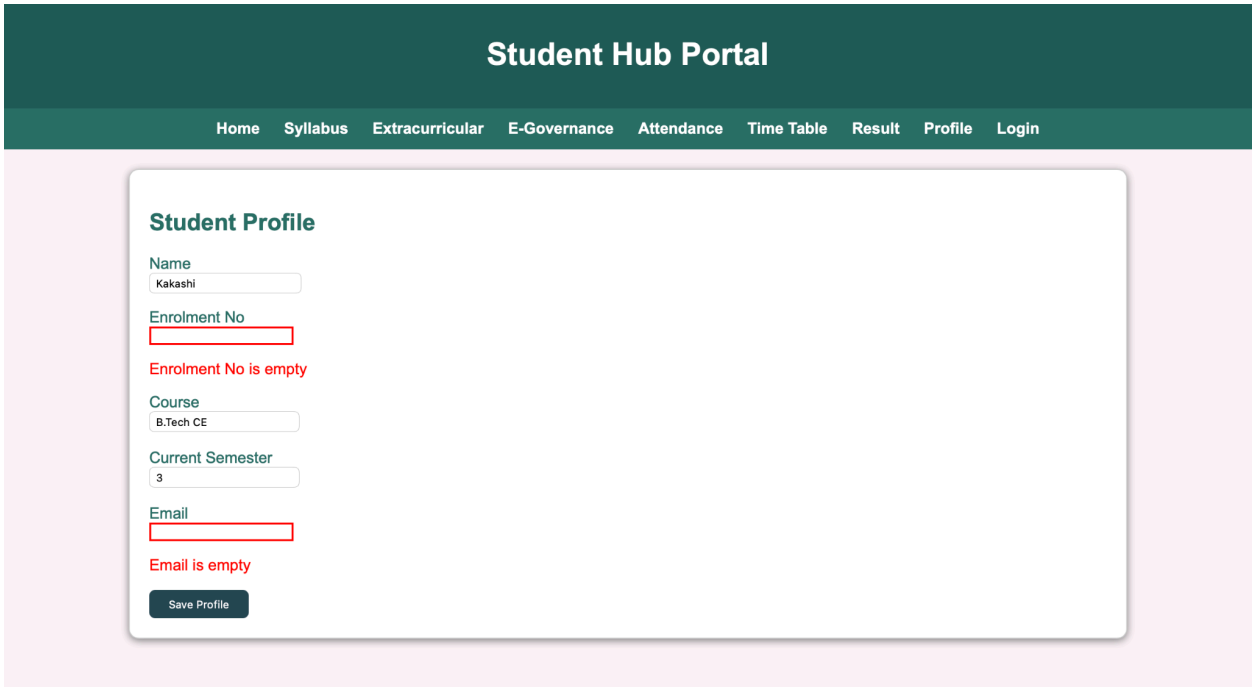


Figure 4.2 : Profile page (peofile.html)

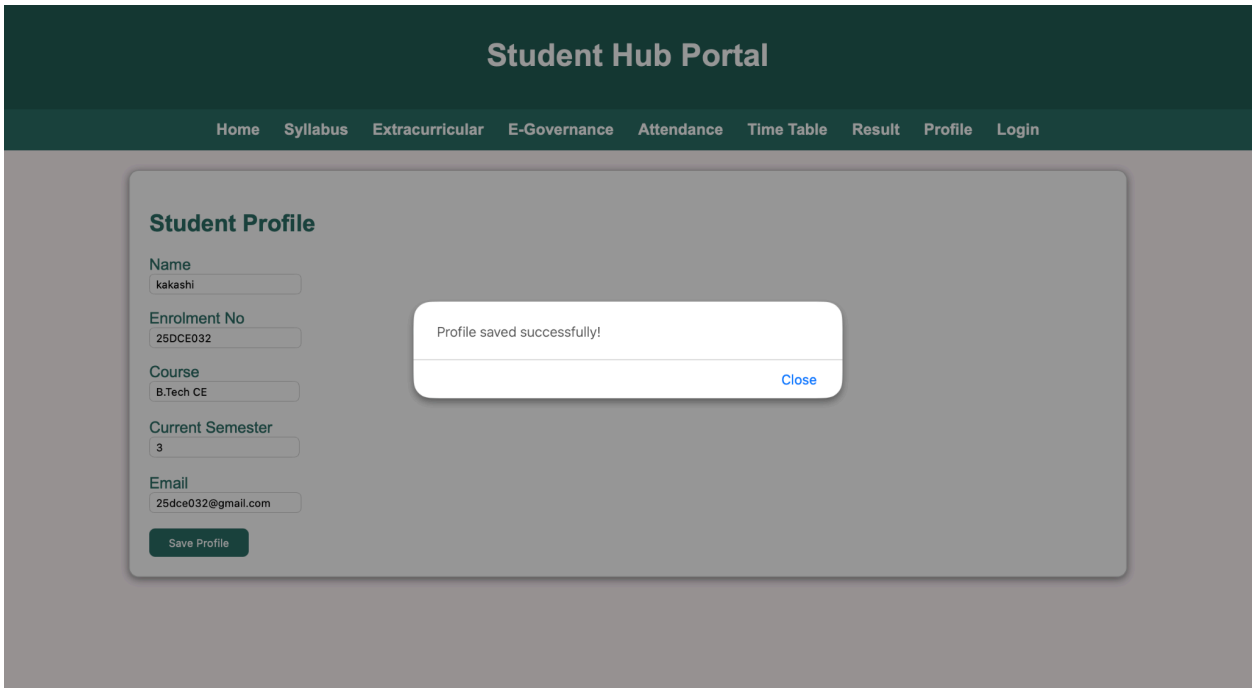
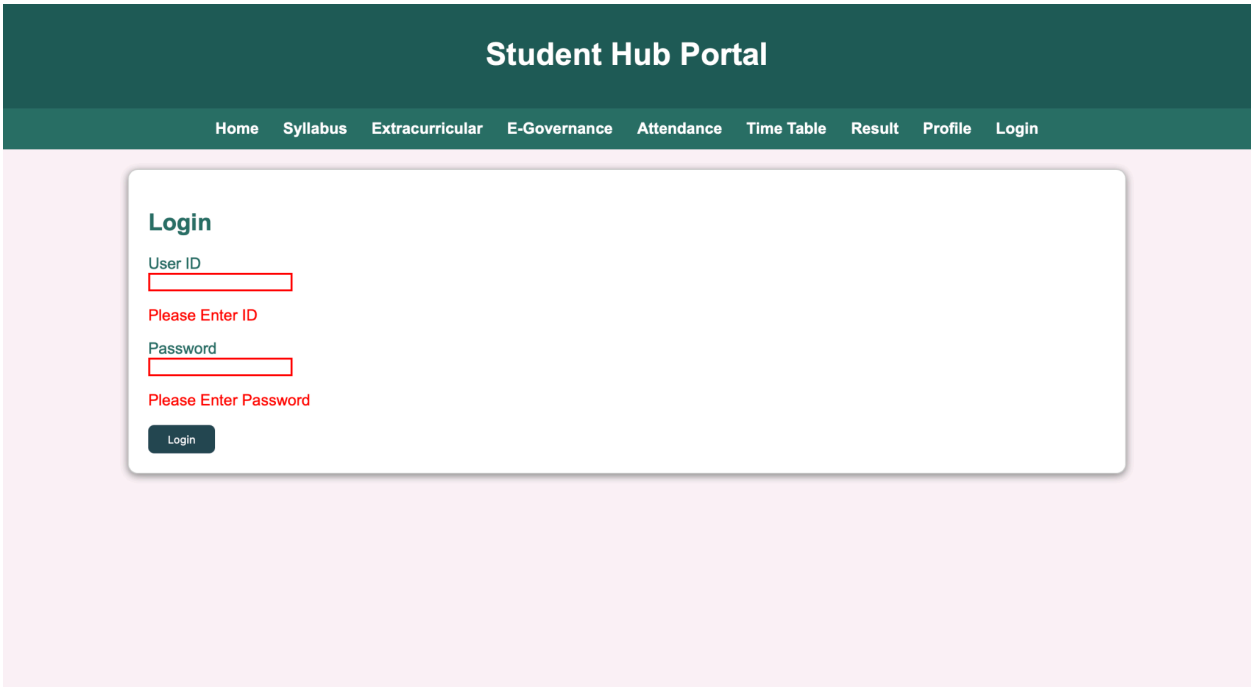


Figure 4.3: Login page (login.html)



D. GitHub Repository Link:

https://github.com/dhyaani19/StudentHub-Portal.git

E. Challenges Faced:

While doing this practical, I faced some difficulties in understanding how to validate user input using JavaScript. Creating the correct Regex for different fields was a little confusing at first. I also had some problems showing the error message at the correct place and changing the input box style when the data was invalid. Sometimes the validation worked for one field but not for another. I solved these problems by checking my code carefully and testing each input field one by one.

F. Learning Outcomes:

n this practical, I learned how to take input from the user using JavaScript and validate the entered data. I learned how to check if a box is empty and show an error message when the input is invalid. I also learned how to use Regular Expressions (Regex) to validate different fields in the Login and Profile forms. I learned how to change the input box style when the entered data is incorrect. Overall, I understood how JavaScript can be used to make forms more interactive and user-friendly.

Faculty Remarks: